

Corporate Programs - Code Explanation (Hinglish)

Yeh folder ek "Corporate Sponsored Degree & Training Registration System" implement karta hai. Isme alag-alag companies (jaise TCS, Wipro) apne employees ya candidates ke liye sponsored programs (MCA, M.Tech) run karti hain, jisme eligibility aur selection criteria apply hota hai.

Sir ko impress karne ke liye main point: Sir ko batana ki yeh ek Real-world problem ka **Object-Oriented Design (OOPs)** hai. Isme humne **Inheritance** (Base class se child class banana) aur **Smart Pointers** (**shared_ptr**) ka use kiya hai jisse memory leak na ho, aur pointers safely khud delete ho jayein.

1. AcademicRecord.h

Yeh class candidate ki academic details store karti hai. (Jaise marksheet)

```
#pragma once
#include <string>

using namespace std;

class AcademicRecord {
    string degree; // Kounsi degree ki hai (e.g., BCA, BTECH)
    double cgpa;    // Kitne marks hain
    int graduationYear; // Kab pass kiya

public:
    // Constructor: Jab naya record banega, toh yeh sari details initialize karega
    AcademicRecord(string degree, double cgpa, int graduationYear)
        : degree(degree), cgpa(cgpa), graduationYear(graduationYear) {}

    // Getters: Bahar se in values ko read karne ke functions
    double getCgpa() const { return cgpa; }
    string getDegree() const { return degree; }
    int getGraduationYear() const { return graduationYear; }
};
```

2. Candidate.h

Candidate class ek applicant ki profile maintain karti hai. Ek bachha jab apply karta hai toh wo apni detail aur apni marksheet (AcademicRecord) le kar aata hai.

```
#pragma once
#include <string>
#include "AcademicRecord.h"

using namespace std;
```

```

class Candidate {
    string name;
    string email;
    AcademicRecord record; // Candidate ka academic data store ho raha hai
    (Composition concept)

public:
    // Constructor
    Candidate(string name, string email, AcademicRecord record)
        : name(name), email(email), record(record) {}

    string getName() const { return name; }
    // Const reference se record return kar raha hai taaki data copy na ho
    (performance ke liye)
    const AcademicRecord& getRecord() const { return record; }
    string getEmail() const { return email; }
};

```

3. CorporateProgram.h

Yeh class corporate program ki details define karti hai, jaise TCS Ignite ya Wipro WILP.

```

#pragma once
#include <string>

using namespace std;

class CorporateProgram {
    string programName; // Jaise: "WILP M.Tech"
    string category;     // Jaise: "MTECH"
    int durationMonths;  // Program kitne mahine ka hai

public:
    CorporateProgram(string programName, string category, int
durationMonths)
        : programName(programName), category(category),
durationMonths(durationMonths) {}

    string getProgramName() const { return programName; }
    string getCategory() const { return category; }
    int getDurationMonths() const { return durationMonths; }
};

```

4. Company.h

Company class (e.g., TCS) programs ki ek list maintain karti hai jo woh offer kar rahi hai.

```

#pragma once
#include <string>

```

```
#include <vector>
#include <memory>
#include "CorporateProgram.h"

using namespace std;

class Company {
    string name;
    // 'vector' ek dynamic array hai.
    // 'shared_ptr' ka use isliye kiya hai taaki memory khud manage ho.
    // Agar program ka use khatam ho gaya toh memory delete ho jayegi.
    vector<shared_ptr<CorporateProgram>> offeredPrograms;

public:
    Company(string name) : name(name) {}

    // Ek naya program company me add karne ke liye
    void addProgram(shared_ptr<CorporateProgram> program) {
        offeredPrograms.push_back(program);
    }

    string getName() const { return name; }
};
```

5. SelectionCriteria.h

Yeh ek **abstract base class (interface)** hai. Yeh bahut important hai. Isme sirf rules ke naam hain, rules actually kaise kaam karenge yeh dusri class batayegi.

```
#pragma once
#include "Candidate.h"

class SelectionCriteria {
public:
    // '= 0' ka matlab yeh "Pure Virtual Function" hai.
    // Jo bhi is class ko inherit karega, use yeh function likhne padenge.
    virtual bool isEligible(const Candidate& candidate) const = 0; //
    Eligible hai ya nahi
    virtual double calculateScore(const Candidate& candidate) const = 0; //
    Score kitna hai
    virtual ~SelectionCriteria() = default; // Destructor
};
```

6. StandardSelectionCriteria.h

Yeh class purani class ([SelectionCriteria](#)) ko inherit karti hai aur asal logic likhti hai.

```
#pragma once
#include <vector>
```

```

#include <string>
#include <algorithm>
#include "SelectionCriteria.h"

using namespace std;

// 'public SelectionCriteria' matlab isne us interface ko inherit kiya
class StandardSelectionCriteria : public SelectionCriteria {
    double minCgpa; // Minimum required CGPA
    vector<string> allowedDegrees; // Allowed degrees ki list (jaise BCA,
    BSC sirf)

public:
    StandardSelectionCriteria(double minCgpa, const vector<string>&
    allowedDegrees)
        : minCgpa(minCgpa), allowedDegrees(allowedDegrees) {}

    // isEligible ko override kiya (apna rule likha)
    bool isEligible(const Candidate& candidate) const override {
        const AcademicRecord& rec = candidate.getRecord();

        // Find function check karega ki bacche ki degree allowedDegrees me
        hai ya nahi
        bool degreeAllowed = find(allowedDegrees.begin(),
        allowedDegrees.end(), rec.getDegree()) != allowedDegrees.end();

        // Agar marks limit se zyada hain AND degree allowed hai, tabhi
        TRUE hoga.
        return rec.getCgpa() >= minCgpa && degreeAllowed;
    }

    // Score calculate karne ka rule (yahan simply CGPA return kar rahe
    hain)
    double calculateScore(const Candidate& candidate) const override {
        return candidate.getRecord().getCgpa();
    }
};

```

7. RecruitmentProcess.h

Yeh poore process ka engine hai. Forms receive karta hai aur final list nikalta hai.

```

#pragma once
#include <iostream>
#include <vector>
#include <memory>
#include <algorithm>
#include "Company.h"
#include "CorporateProgram.h"
#include "SelectionCriteria.h"
#include "Candidate.h"

```

```

using namespace std;

class RecruitmentProcess {
    shared_ptr<Company> company;
    shared_ptr<CorporateProgram> program;
    int batchYear;
    int intakeCapacity; // Sirf kitne bacche lene hain (Seats)
    shared_ptr<SelectionCriteria> criteria; // Selection ka rule
    vector<Candidate*> applicants; // Jinhone apply kiya
    vector<Candidate*> selectedCandidates; // Jo pass hue

public:
    // Process shuru karne se pehle sari zaruri cheezein bataani padengi
    RecruitmentProcess(shared_ptr<Company> company,
        shared_ptr<CorporateProgram> program, int batchYear, int intakeCapacity,
        shared_ptr<SelectionCriteria> criteria)
        : company(company), program(program), batchYear(batchYear),
        intakeCapacity(intakeCapacity), criteria(criteria) {}

    // Bachha jab form bharega
    void apply(Candidate& candidate) {
        applicants.push_back(&candidate);
    }

    // Pura selection chalane ka function
    void processSelection() {
        vector<Candidate*> eligible;

        // 1. Eligibility check karo
        for (Candidate* c : applicants) {
            // Hamara banaya hua rule isEligible check hoga
            if (criteria->isEligible(*c)) {
                eligible.push_back(c);
            }
        }

        // 2. Sorting karo (Jiske CGPA sabse jyada, wo sabse upar)
        // Lamda function ka use karke descending order me sort kar rahe
        // hain
        sort(eligible.begin(), eligible.end(), [this](Candidate* c1,
            Candidate* c2) {
                return criteria->calculateScore(*c1) > criteria-
                >calculateScore(*c2);
            });

        cout << "--- Selection List for " << company->getName() << " " <<
            program->getProgramName() << " (" << batchYear << " ) ---" << endl;
        int count = 0;

        // 3. Intake capacity (seats) ke hisab se baccho ko finally select
        // karo
        for (Candidate* c : eligible) {
            if (count < intakeCapacity) {

```

```

        selectedCandidates.push_back(c);
        cout << c->getName() << " - Selected with CGPA: " << c-
>getRecord().getCgpa() << endl;
        count++; // Ek seat bhar gayi
    }
}

if (selectedCandidates.empty()) {
    cout << "No candidates met the eligibility criteria or
applied." << endl;
}
cout << "-----" <<
endl;
}
};

```

8. main.cpp

Main program jahan saari classes jud ke run hoti hain.

```

#include <iostream>
#include <memory>
#include <vector>
#include <string>
#include "AcademicRecord.h"
#include "Candidate.h"
#include "SelectionCriteria.h"
#include "StandardSelectionCriteria.h"
#include "CorporateProgram.h"
#include "Company.h"
#include "RecruitmentProcess.h"

using namespace std;

int main() {
    cout << "=== Corporate Sponsored Degree & Training Registration System
===\n" << endl;

    // 1. Companies create ki (TCS aur Wipro)
    auto tcs = make_shared<Company>("TCS");
    auto wipro = make_shared<Company>("Wipro");

    // 2. Corporate programs banaye
    auto tcsIgnite = make_shared<CorporateProgram>("Ignite MCA", "MCA",
24);
    auto wiproWilp = make_shared<CorporateProgram>("WILP M.Tech", "MTECH",
48);

    // Company ke andar program add kiya
    tcs->addProgram(tcsIgnite);
    wipro->addProgram(wiproWilp);
}

```

```
// 3. Bachhe aur unke records banaye
AcademicRecord rec1("BCA", 8.5, 2024);
Candidate c1("Ramesh", "ramesh@email.com", rec1); // BCA wala

AcademicRecord rec2("BSC", 9.2, 2024);
Candidate c2("Suresh", "suresh@email.com", rec2); // BSC wala

AcademicRecord rec3("BTECH", 7.5, 2024);
Candidate c3("Anita", "anita@email.com", rec3); // BTECH wali

// 4. Criteria set kiya
// Wipro ko BCA, BSC wale minimum 6.0 CGPA chahiye
auto wilpCriteria = make_shared<StandardSelectionCriteria>(6.0,
vector<string>{"BCA", "BSC"});
// TCS ko BCA, BSC wale minimum 7.0 CGPA chahiye
auto igniteCriteria = make_shared<StandardSelectionCriteria>(7.0,
vector<string>{"BCA", "BSC"});

// 5. Recruitment processes initialize kiye
// Wipro process me sirf 1 seat hai (intakeCapacity = 1)
RecruitmentProcess wiproProcess(wipro, wiproWilp, 2024, 1,
wilpCriteria);
// TCS process me 2 seat hain (intakeCapacity = 2)
RecruitmentProcess tcsProcess(tcs, tcsIgnite, 2024, 2, igniteCriteria);

// 6. Bachho ne apply kiya
wiproProcess.apply(c1);
wiproProcess.apply(c2);
wiproProcess.apply(c3); // Anita reject hogi kyonki uska BTECH hai jo
list me allow nahi hai

tcsProcess.apply(c1);
tcsProcess.apply(c2);

// 7. Process chalana aur list nikalna
wiproProcess.processSelection();
// Suresh aage hoga list me kyonki uske 9.2 CGPA hai (Seat 1 hi hai toh
sirf Suresh select hoga)
cout << endl;

tcsProcess.processSelection();
// Yahan 2 seats hain toh Ramesh aur Suresh dono select honge

return 0;
}
```